

Samuel Balent

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EDUCATION

University of Colorado Boulder

Boulder, CO

Bachelor in Arts Computer Science; Bachelor in Arts Economics, Minor in Mathematics

Expected May 2028

- **Leadership/ Clubs:** Head of Engineering at CU Quant; Pledge Team at Kappa Theta Pi (Professional Technology Fraternity), HackCU, Quantum Scholar, Blockchain Club, Jellystone Activities Team Lead

EXPERIENCE

Software Engineer/ Applied AI Intern

Remote

Morrison Street Advisors

Aug 2025 – Present

- Engineered GPT-4 chat-bot with custom prompt engineering to prioritize internal endpoint searches and prevent prompt injection attacks, integrating custom sitemap endpoint to serve context-aware platform resources.
- Built Rails middle ware for API-Sports integration with 5-minute cache layer, reducing API calls by 80improving response times from 2s to 200ms for frequently accessed data.
- Evaluated 4 AI tools (Delphi, OpenAI Whisper, Pomelli, Opus) through technical prototyping and performance benchmarking, delivering executive reports that informed 150K+ investment decisions.

Head of Algorithmic Trading

Boulder, CO

CU Quants

Aug 2025 – Present

- Lead 6-person engineering team building algorithmic trading infrastructure that generated 39.5% annualized returns (9.74% quarterly) across 14 stablecoin markets.
- Designed and executed market-making strategies in illiquid stablecoins using the in-house SpeedByte and XLRTS platform. Supervised daily trade execution and risk oversight, with 23,000+ trades executed and no down days.
- Designed and implemented risk management framework using VaR and ES models to monitor tail-risk exposure in real-time, reducing draw down events by 30%.

Course Assistant

Boulder, CO

University Of Colorado Boulder

Aug 2025 – Present

- Mentored 200+ students in Data Structures and Python fundamentals, debugging, and algorithmic problem-solving, increasing assignment completion rates by 15% through targeted office hours and code reviews.
- Developed visual learning aids and hands-on exercises for CS and non-CS majors, improving comprehension rates by 20% as measured by pre/post-assessment scores.

PROJECTS

SpeedByte | Python, React, WebSockets

Feb 2026

- Led a team of 6 engineers to architect and ship a real-time algorithmic stablecoin trading platform in Python.
- Engineered a Python SDK for the Gemini exchange with full REST and WebSocket support, enabling seamless retrieval of account, market, and trading data.
- Designed a dynamic position sizing engine that adjusts trade exposure in real time based on live market conditions across 3 exchanges (Kraken, OKX, Gemini) via WebSocket streams.

Real-Time Face Detection System | Python, TensorFlow, OpenCV, Albumentations

Dec 2026

- Developed end-to-end face detection pipeline with custom VGG16-based CNN architecture, achieving dual-head classification and bounding box regression for live webcam inference.
- Engineered automated data augmentation pipeline generating 7,200 training samples from 120 base images with dynamic resizing, implementing custom TensorFlow loss functions.

TECHNICAL SKILLS AND INTEREST

Languages: Python, C++, C, Java, JavaScript, Ruby, TypeScript, SQL, HTML/CSS

Frameworks & Libraries: Pandas, NumPy, React.js, Node.js, Express.js, FastAPI, Swing, Ruby on Rails

Tools & Platforms: Git, GitHub, Bit Bucket, Docker, PostgreSQL, Supabase, Jira, VS Code, Kali Linux

Core Competencies: Data Structures & Algorithms, Parallel Computing, Quantitative Finance, Risk Modeling (VaR/ES), LLM Integration, Agile/Scrum